

VTU Belagavi

MID-TERM EXAMINATION ? 2025

Geotechnical Engineering ? Mid-Term Examination, 2025

Subject Code: CE401 Semester: 4 Regulation: R18

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Soil Properties and Classification)

1. Solve a problem related to Soil Properties and Classification. Show all the steps clearly. [12 Marks]
2. Write short notes on Soil Properties and Classification. [12 Marks]
3. List the advantages and limitations of Soil Properties and Classification. Explain any two in detail. [8 Marks]

SECTION B (Unit 2: Permeability and Seepage)

4. Describe the working principle of Permeability and Seepage with necessary equations. [12 Marks]
5. List the advantages and limitations of Permeability and Seepage. Explain any two in detail. [12 Marks]
6. Explain Permeability and Seepage with a neat diagram/sketch and a suitable example. [12 Marks]

SECTION C (Unit 3: Compaction and Consolidation)

7. Compare and contrast the key concepts involved in Compaction and Consolidation. [8 Marks]
8. Solve a problem related to Compaction and Consolidation. Show all the steps clearly. [12 Marks]
9. Explain Compaction and Consolidation with a neat diagram/sketch and a suitable example. [12 Marks]

SECTION D (Unit 4: Shear Strength of Soils)

10. Write short notes on Shear Strength of Soils. [8 Marks]
11. Discuss the engineering applications of Shear Strength of Soils. [12 Marks]
12. Write short notes on Shear Strength of Soils. [12 Marks]

SECTION E (Unit 5: Bearing Capacity and Settlement)

13. List the advantages and limitations of Bearing Capacity and Settlement. Explain any two in detail. [12 Marks]
14. Write short notes on Bearing Capacity and Settlement. [8 Marks]
15. Discuss the engineering applications of Bearing Capacity and Settlement. [12 Marks]

Marks]

*** End of Question Paper ***